

Education

Bachelor of Technology – Computer Science with Artificial Intelligence

May 2022 – May 2026

Senior Secondary Education (Class XII – CBSE)

2019 – 2021

Secondary Education (Class X – CBSE)

2019

Experience

Software Development Intern – SaaS Platform

June 2025 – Dec 2025

- Refactored a **monolithic engineering workspace** into a **modular SaaS architecture** for structural analysis applications.
 - Reduced component complexity by **76%** using modern frontend frameworks and state management tools.
 - Implemented **3D visualization workflows** for loads, supports, structural responses, and section configurations using WebGL-based rendering libraries.
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AI Research Intern

June 2023 – July 2023

- Contributed to research on **medical imaging for pancreatic cancer detection** using CNN, RNN, and object detection techniques.
 - Evaluated and annotated **150+ CT scans** of abdominal regions for training datasets.
 - Helped improve diagnostic workflows and reduce clinical analysis time significantly.
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Student Tech Community – Operations Lead

Nov 2023 – Present

- Managed operations for a student developer community with **45+ members**.
 - Organized hackathons, workshops, and industry talks on **deep learning, machine learning, and diffusion models**.
 - Increased community engagement across developer platforms and communication channels.
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Publications

Adaptive Swarm Optimization Algorithm

- Proposed improvements to an existing swarm optimization algorithm with **adaptive step-size control and enhanced exploration strategies**.
 - Achieved improved performance on **60 out of 81 benchmark functions** from standard optimization test suites.
 - Submitted to a peer-reviewed engineering software journal.
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Explainable IIoT Security Framework

- Developed an **Explainable AI-based intrusion detection system** for Industrial IoT networks.
 - Integrated explanation techniques such as **SHAP, Integrated Gradients, and DeepLIFT**.
 - Optimized edge deployment using pruning and model quantization, achieving **up to 74% model size reduction**.
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Projects

IoT Network Anomaly Detection System

Jan 2024

- Built a machine learning system to detect **device spoofing, man-in-the-middle attacks, and DDoS attacks** in IoT networks.
 - Generated synthetic traffic data using IoT simulators and analyzed packet-level behavior.
 - Used machine learning models to classify anomalous network activity.
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Skills

Programming Languages

C/C++, Python, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Tools

React, Node.js, Git, Pandas, NumPy, Matplotlib, TensorFlow, PyTorch, OpenCV, Scikit-learn, Power BI

Design & Development Tools

UI/UX design tools, version control platforms

Networking & Systems

OSI Model, TCP/IP, DHCP, DNS, IPv4/IPv6, RAID